

An Expert-Following Strategy for Polymarket

Methodology and Backtest Results

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Executive Summary

- We test a settlement-only Polymarket strategy that follows historically skilled traders.
- Trader expertise is estimated from past settled markets using semantic similarity between market questions.
- The current strategy trades only the high-probability “stable” bucket: entry price > 0.8 .
- The latest full-market-scale backtest covers roughly 3.11 years of trade history.
- Final portfolio value is evaluated using total equity, not only cash balance.

Data Sources and Coverage

- Market metadata is fetched from the Polymarket Gamma API.
- Public historical trades are fetched from the Polymarket Data API.
- Cached full-market run:
 - 299,318 market records.
 - 1,195,104 normalized trade rows.
 - 2,993 trade-covered markets.
 - 2,516 trade-covered markets with available resolution labels.
- Local cache is used for markets, trades, and embeddings to make repeated runs practical.

Important Modeling Assumption

Settlement-only PnL

We do not model mid-market exits, stop-losses, or selling before resolution. Every position is evaluated by final settlement or by mark-to-last-known trade price if still open.

- This differs from stock-style backtests.
- It matches the intended research question: buy a prediction market outcome and hold until final resolution.
- Reported `total_equity` is preferred over `final_balance`, because open positions may lock cash.

Anti-Leakage Design

- The event stream is processed chronologically.
- At time t , trader skill uses only markets settled no later than t .
- Current market resolution is never used to generate current signals.
- A signal is executed only after a configurable delay.
- Time-to-resolution filters are computed using market metadata available at entry time.

Trader Market Score

For trader u in settled market m :

$$\text{score}_{u,m} = \text{clip} \left(\frac{\text{PnL}_{u,m}}{\text{notional}_{u,m}}, -1, 1 \right)$$

where:

- `notional` is absolute traded dollar volume.
- `PnL` is computed from cash flows plus final settlement payout.
- Scores are clipped to reduce extreme unit-return outliers.

Semantic Similarity Between Markets

Each market question is embedded into a vector:

$$e_m = \text{Embed}(\text{question}_m)$$

Similarity is cosine similarity:

$$s(m, n) = \max(0, \cos(e_m, e_n))$$

Current implementation supports:

- `sentence_transformers`: currently BAAI/bge-large-en-v1.5.
- `hashing`: faster fallback for parameter sweeps.

Trader Skill on a New Market

For trader u and target market q at time t :

$$\widehat{\text{skill}}_{u,q,t} = \frac{\sum_{m \in \mathcal{H}_{u,t}} s(q, m) \cdot \text{notional}_{u,m} \cdot \text{score}_{u,m}}{\sum_{m \in \mathcal{H}_{u,t}} s(q, m) \cdot \text{notional}_{u,m}}$$

where $\mathcal{H}_{u,t}$ contains only markets already settled by time t .

A trader is considered skilled in the current flow if:

$$\widehat{\text{skill}}_{u,q,t} \geq \text{skill_threshold}$$

Direction Consensus Signal

Each skilled trade contributes direction-weighted flow:

$$w_i = \widehat{\text{skill}}_i \cdot \text{volume}_i$$

Direction ratios:

$$R_{\text{YES}} = \frac{\sum_{i \in \text{YES}} w_i}{\sum_i w_i}, \quad R_{\text{NO}} = \frac{\sum_{i \in \text{NO}} w_i}{\sum_i w_i}$$

A signal is triggered when:

$$\max(R_{\text{YES}}, R_{\text{NO}}) \geq \text{consensus_threshold}$$

Current Strategy Rules

Baseline fixed-fraction configuration:

- Trade only the stable bucket: entry token price > 0.8 .
- Lottery bucket is disabled.
- Direction must pass expert consensus threshold.
- Entry must satisfy:

$$5 < \text{days to resolution} < 40$$

- Stable bucket position size:

$$\text{notional} = 0.2 \times \text{current cash balance}$$

- Maximum two entries per market.

Target Exposure Sizing

Alternative sizing experiment:

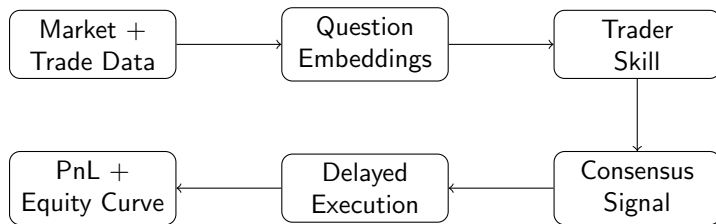
$$\text{target deployed} = 0.97 \times \text{total equity}$$

$$\text{available cash} = \text{cash} - 0.05 \times \text{total equity}$$

$$\text{annualized edge} = \max(\text{confidence} - \text{entry price}, 0) \times \frac{365}{\text{days to resolution}}$$

$$\text{order notional} = \text{remaining target budget} \times \text{clip}(0.25 \times \text{annualized edge}, 0.05, 0.35)$$

Backtest Framework in One Slide



Backtest Result Snapshot

Metric	Fixed 20%	Target Exposure
Trade period	2023-03-16 to 2026-04-24	
Duration	1134.95 days / 3.107 years	
Markets in universe	299,318	
Historical trades sampled	1,195,104	
Unique traders	330,576	
Strategy trades	513	514
Win rate	99.03%	99.03%
Total realized PnL	12,541.45	22,783.38
Average PnL per settled trade	24.45	44.33
Final total equity	22,600.15	33,220.86
Approx. CAGR	30.00%	47.12%

Realized Results Comparison

Metric	Fixed 20%	Target Exposure
trades	513	514
win_rate	0.990253	0.990272
total_pnl	12,541.4474	22,783.3788
avg_pnl_per_trade	24.4473	44.3256
pnl_volatility	81.3893	195.9001
final_balance	34.2648	1,589.7532

Portfolio Snapshot Comparison

Metric	Fixed 20%	Target Exposure
open_positions	27	34
open_notional	22,507.1826	31,193.6256
open_market_value	22,565.8838	31,631.1050
open_unrealized_pnl	58.7013	437.4794
cash_balance	34.2648	1,589.7532
total_equity	22,600.1487	33,220.8582

- Target exposure keeps more cash buffer while deploying more total notional.
- The higher ending equity comes with higher realized PnL volatility.

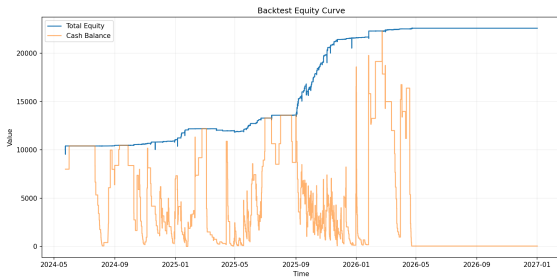
Bucket-Level Results

Bucket	Fixed Trades	Fixed Win Rate	Target Trades	Target Win Rate
Stable: price > 0.8	513	0.990253	514	0.990272

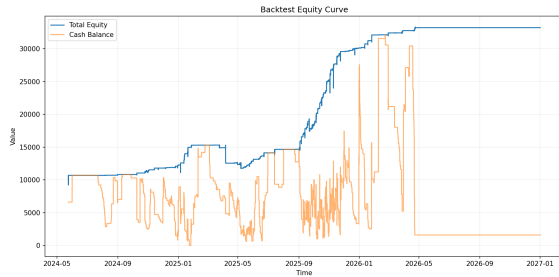
- Both reported configurations trade only the stable bucket.
- The lottery bucket remains disabled in both configurations.

Equity Curve Comparison

Fixed 20% sizing



Target exposure sizing



Result Interpretation

- High win rate comes from trading high-probability stable outcomes.
- Positive total equity indicates that the expert-following filter adds value over this sample.
- Target exposure sizing improves capital deployment and total equity in this run.
- The improvement comes with higher PnL volatility and larger open notional.
- The important portfolio value metric is total equity, not cash alone.

Main Risks and Limitations

- Public API data may be incomplete or schema-shifting.
- Backtest assumes fills at delayed observed trade prices plus slippage.
- Liquidity and market impact are simplified.
- Open positions are marked using last known trade prices.
- Trader skill scores may overstate noisy small-sample performance.
- Parameter choices were research-driven and require out-of-sample validation.

Implemented but Disabled

The codebase includes several research features that are currently turned off in the reported configuration:

- Lottery bucket: trades with $0.1 \leq p \leq 0.5$, controlled by a small fixed lot size.
- Lottery exposure cap: maximum lottery exposure as a fraction of total capital.
- Dynamic entry price cap: stricter or looser maximum entry price based on consensus confidence.
- Minimum edge filter: requires consensus-implied edge over entry price.
- Alternative trader-skill weighting: configurable historical notional exponent.
- Hashing-vectorizer embeddings: fast fallback for parameter sweeps.

Thank You